Report: ETL_Traffic_Data_Analysis

1. Data Preparation

I loaded the collision, parties, victims, and case id datasets using PySpark. After loading the data, I displayed the first five rows from each table to verify that the data was read correctly. Also checked the dataset schemas to confirm columns were correctly formatted and the data types were properly detected.

2. Data Cleaning

2.1. Missing Values

#Check for Missing Values
I checked the datasets for missing values across all columns in the collisions, parties, victims and case_ids tables using PySpark functions. This identified whether any columns contained null values that could affect this analysis. The results showed that the sample dataset did not contain significant missing values, indicating that the data was mostly complete and ready for my further analysis.

#Drop Sparse Columns

I counted the total number of rows first for all, then checked the number of non-null values in each column for the collisions, parties, victims, and case_ids tables. This identified if any columns contained very few values and can be considered sparse. No significantly sparse columns are found in the sample dataset, I removed no columns.

#Convert Data Types

I converted important columns to appropriate data types to ensure accurate analysis. The collision_date column was converted to a date format, and victim_age was converted to integer type. After conversion, the schema is checked to confirm that the columns are correctly formatted.

#Handle Missing Values

I checked the dataset for missing values and decided to handle them using simple imputation. Categorical columns such as weather conditions and lighting are filled with “Unknown”, while numeric values like victim age is filled with 0 where necessary. This approach ensures that useful records are not removed while keeping the dataset consistent for my analysis.

2.2. Fixing Columns

#Remove Duplicates

I checked for duplicate records in all the datasets by comparing the row counts before and after removing duplicates. After applying the duplicate removal step, the counts remained mostly the same, which shows that there are very few or no duplicate records in this sample dataset. This step ensures that the data used for analysis is clean and does not contain many repeated entries.

#Detect Outliers using IQR

I performed outlier detection on relevant numerical columns such as victim age, party age, and collision-related counts. The IQR method is used for age-related variables to identify extreme values, while logical validation is applied to count-based fields such as number of victims. Coordinates are also checked to ensure they fall within valid ranges. This approach ensures that only meaningful numerical fields are analyzed for outliers while avoiding unnecessary processing of categorical or identifier columns.

2.3. Outlier Analysis

I focused only on meaningful numerical columns such as victim age and coordinates. Outlier detection is only relevant for variables where extreme values indicate incorrect or unusual data. Columns like IDs or categorical codes I did not consider.

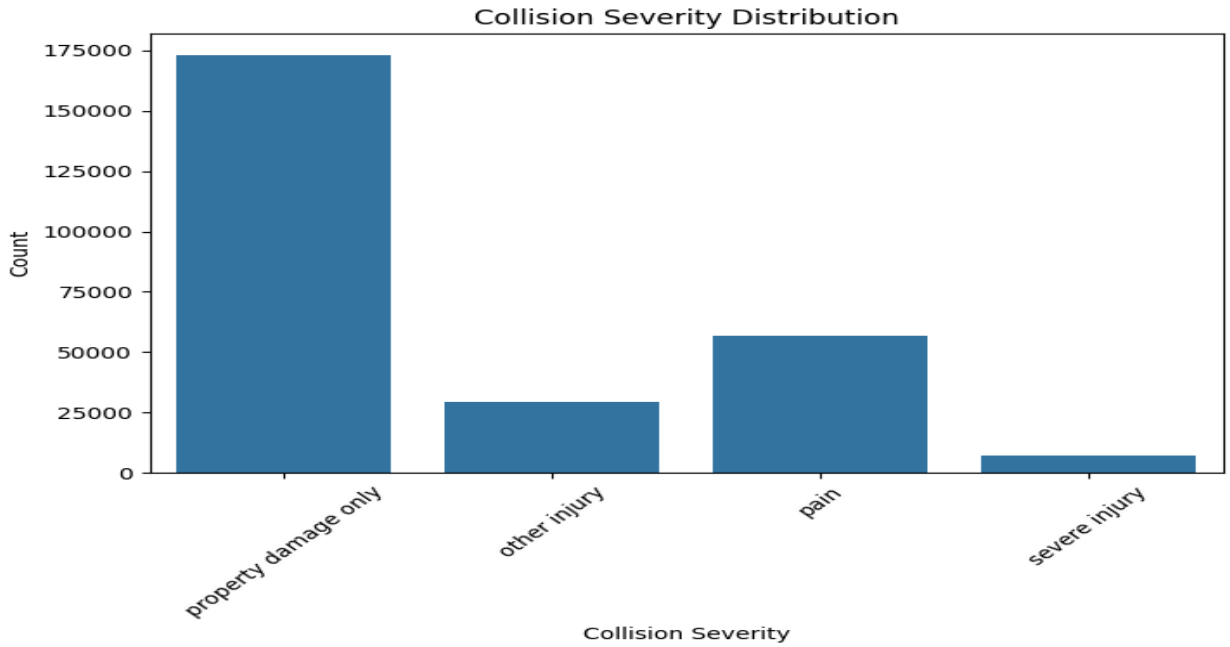
3. Exploratory Data Analysis

3.1.1. Data Preparation

- Columns are selected and arranged in a logical order. Renamed weather, road condition, age, injury level, party type etc column names for better clarity during my further analysis.
- Categorical values are encoded based on the actual categories present in the dataset. The unique values of each column are first examined, and encoding is applied accordingly to ensure correctness and consistency.
- After completing data cleaning steps such as handling missing values, duplicates removing, encoding categorical variables, and correcting data types, the final cleaned dataset are prepared. Then the cleaned data is verified by displaying 5 sample records and checking the schema to ensure consistency and correctness before proceeding to further analysis.

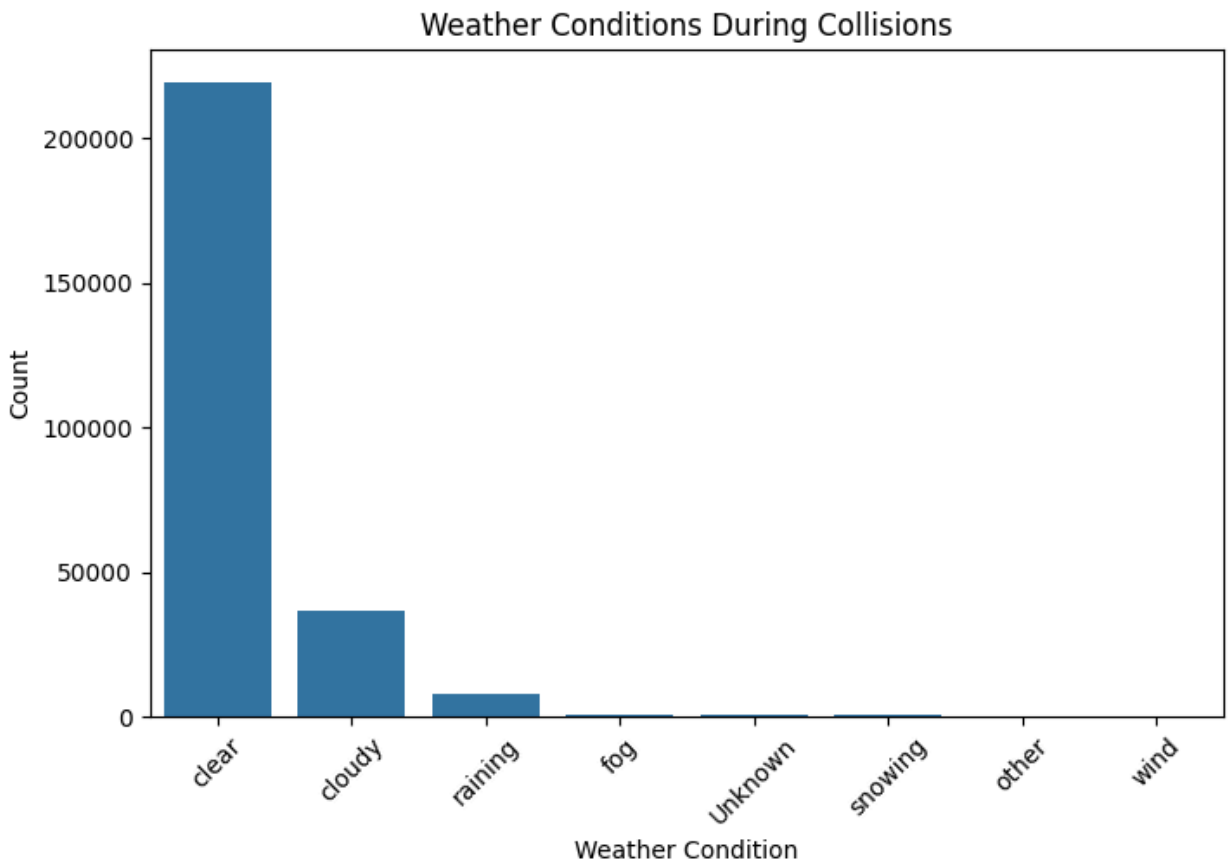
3.1.2. Collision Severity Distribution

I analyzed the distribution of collision severity using a count plot. The results show that most collisions fall under lower severity categories such as property damage only or other injury, while severe occur less frequently. This indicates that although accidents are common, most of them do not result in serious harm.



3.1.3. Weather Conditions During Collisions

I analyzed the weather conditions during collisions using a count plot. The results show that most accidents occur during clear weather conditions, followed by cloudy and rain. This indicates that higher traffic volume during normal weather conditions may contribute more to accidents than extreme weather.

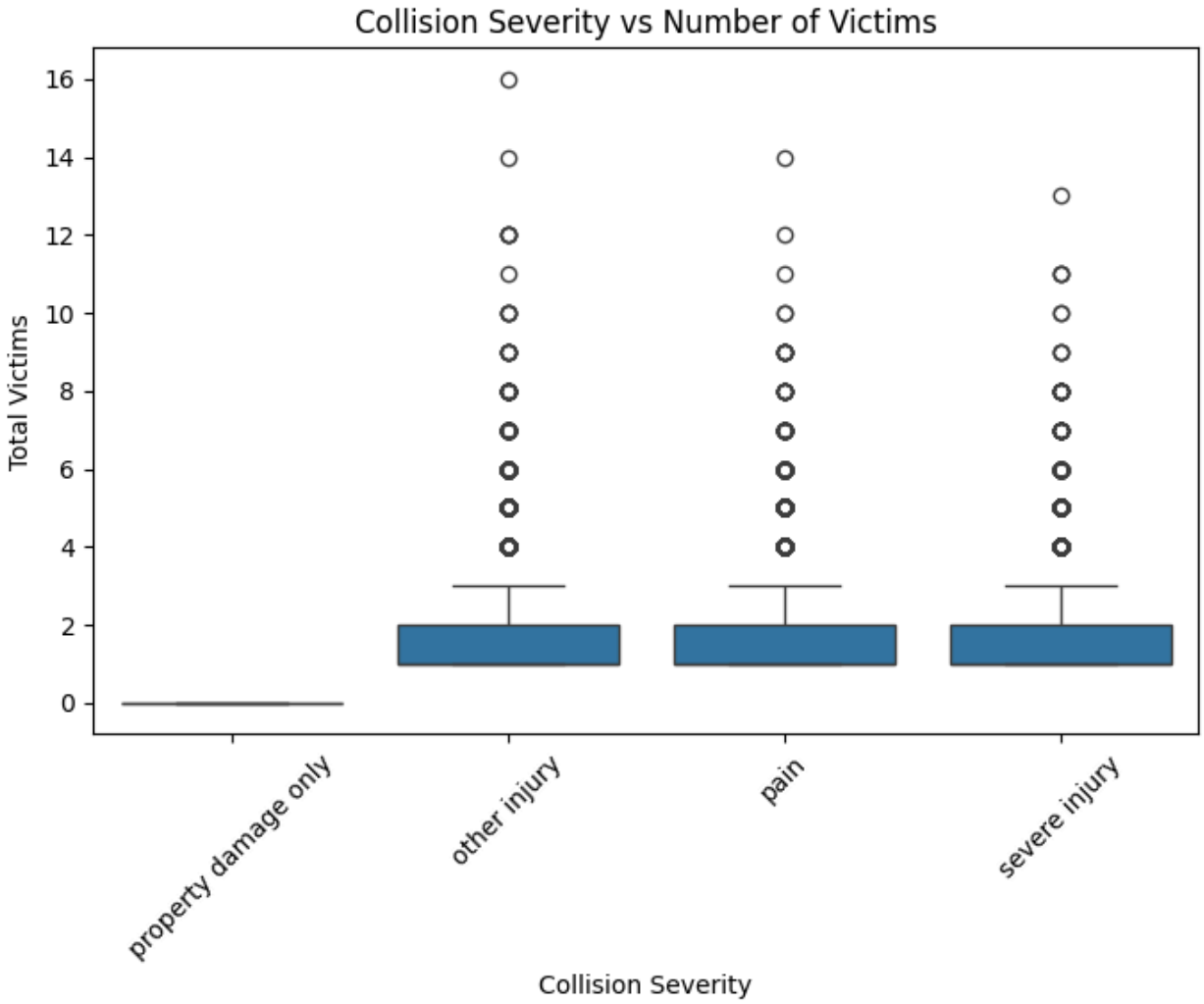


3.1.4. Distribution of Victim Ages

I analyzed the distribution of victim ages using a histogram. The results show that most victims fall within the young to middle-age group, indicating that this age group is more frequently involved in traffic collisions. Very few cases are observed at extreme age ranges.

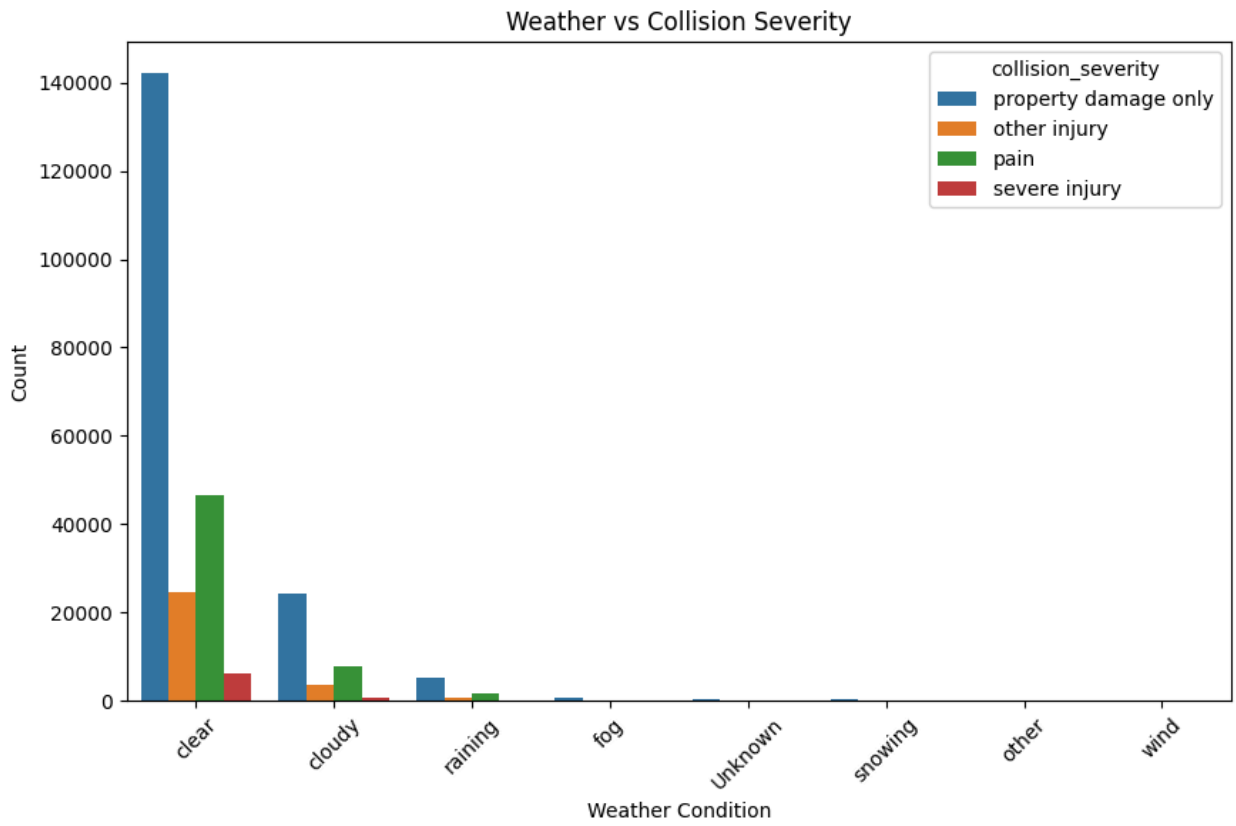
3.1.5. Bivariate Analysis # Collision Severity vs. Number of Victims

I analyzed the relationship between collision severity and the number of victims using a box plot. The results show that other injuries generally involve a greater number of victims. Pain cases tend to have more victims compared to severe collisions.



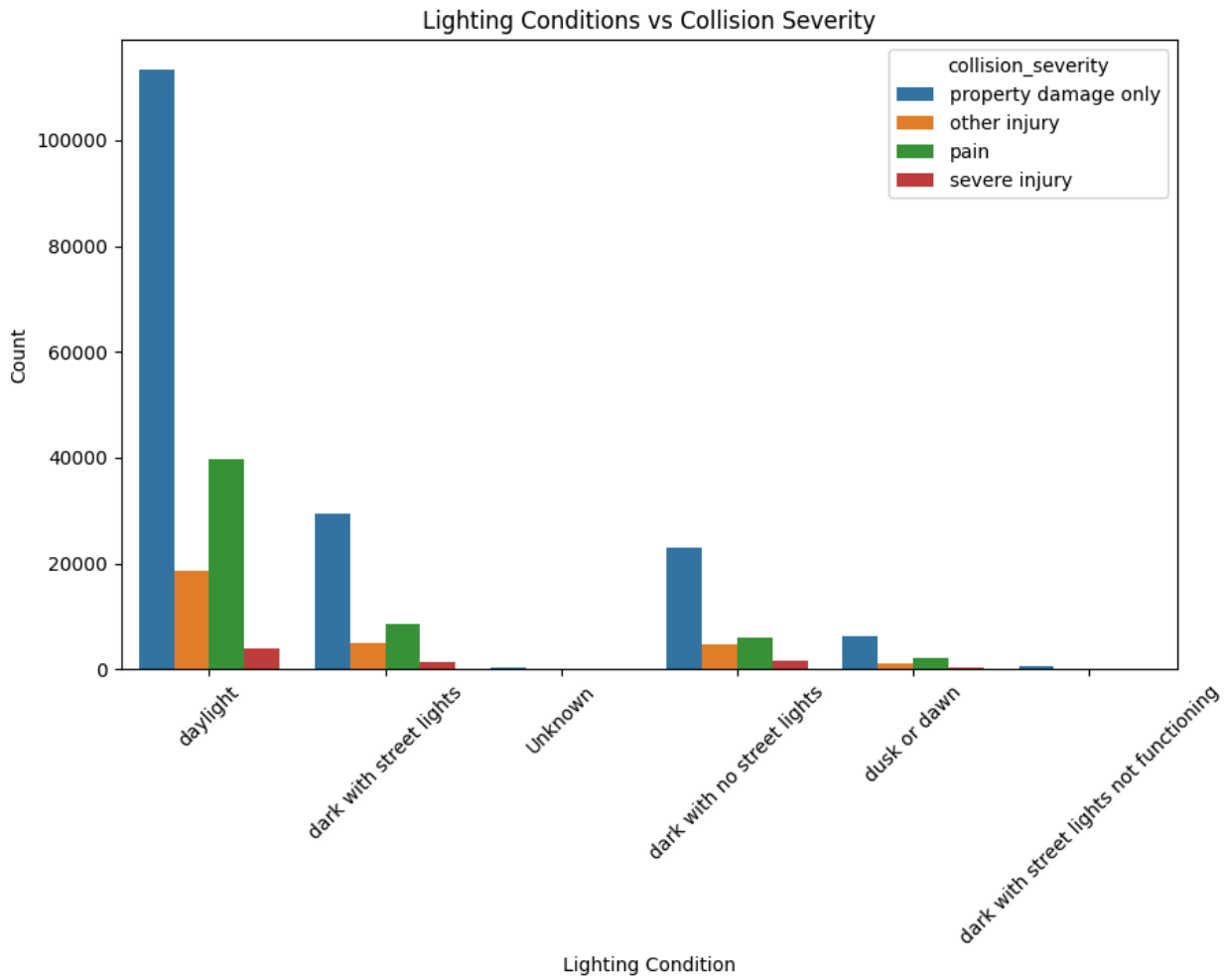
3.1.6. Weather vs. Collision Severity

I analyzed the relationship between weather conditions and collision severity using a grouped bar chart. The results show that most collisions occur during clear weather across all severity levels. Severe and fatal accidents also occur during adverse weather conditions such as cloudy and rain, indicating that weather can also influence the severity of accidents.



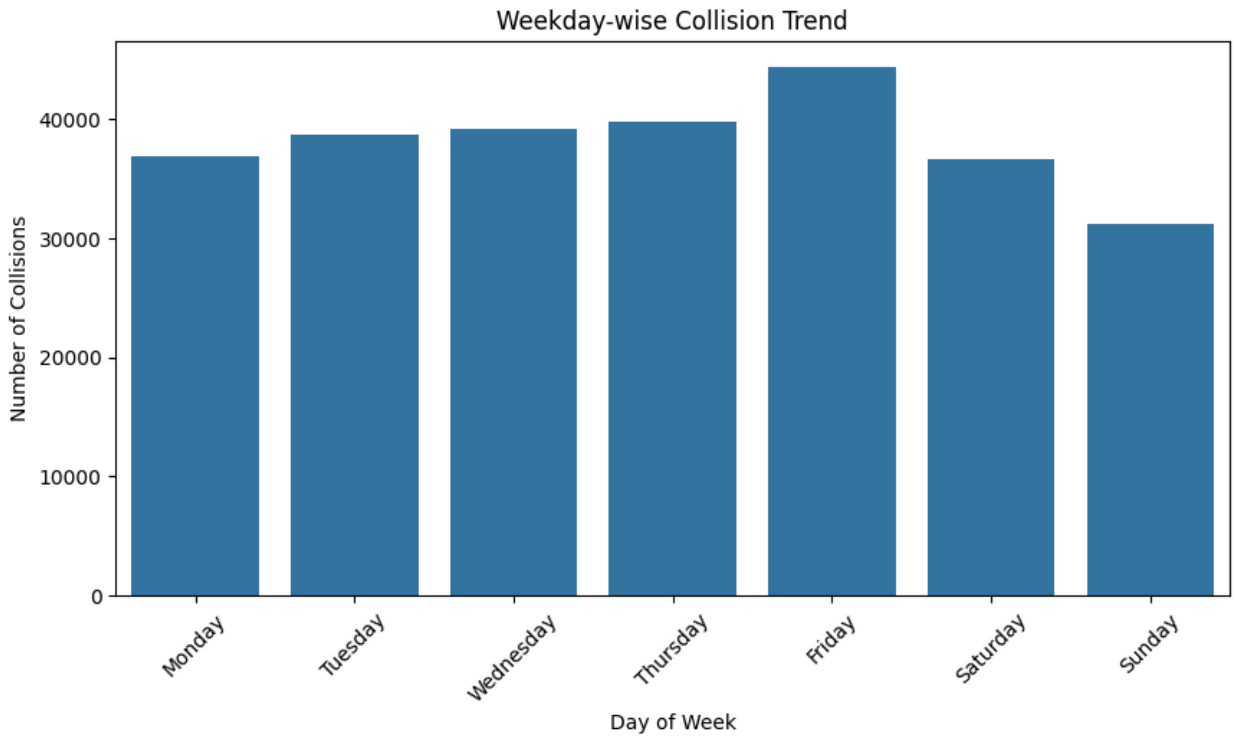
3.1.7. Lighting Conditions vs. Collision Severity

I analyzed the impact of lighting conditions on collision severity using a grouped bar chart. The results show that most accidents occur during daylight conditions due to higher traffic volume. However, severe accidents are also observed during low-light conditions such as in dark with street lights, indicating that poor visibility can increase accident severity.



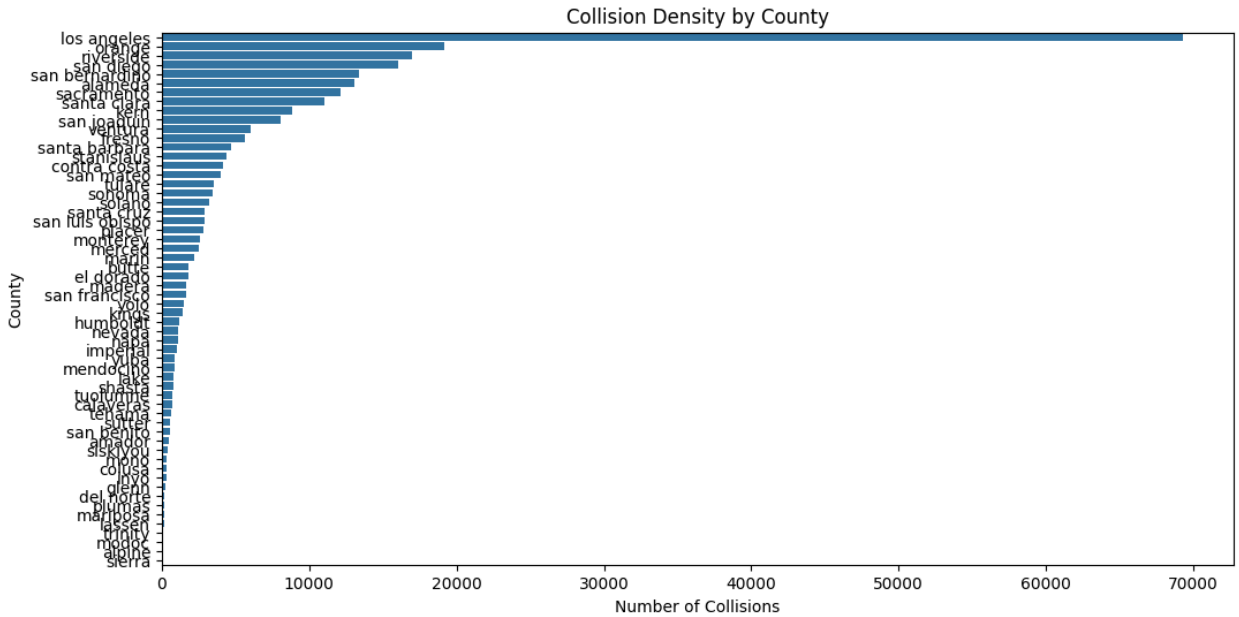
3.1.8. Extract the weekday-wise collision trend

I analyzed collision trends across different days of the week by extracting the weekday from the collision date. The results show that collisions are more frequent on weekdays, especially towards the end of the week, with max on Friday, indicating higher traffic volume and congestion during these days.



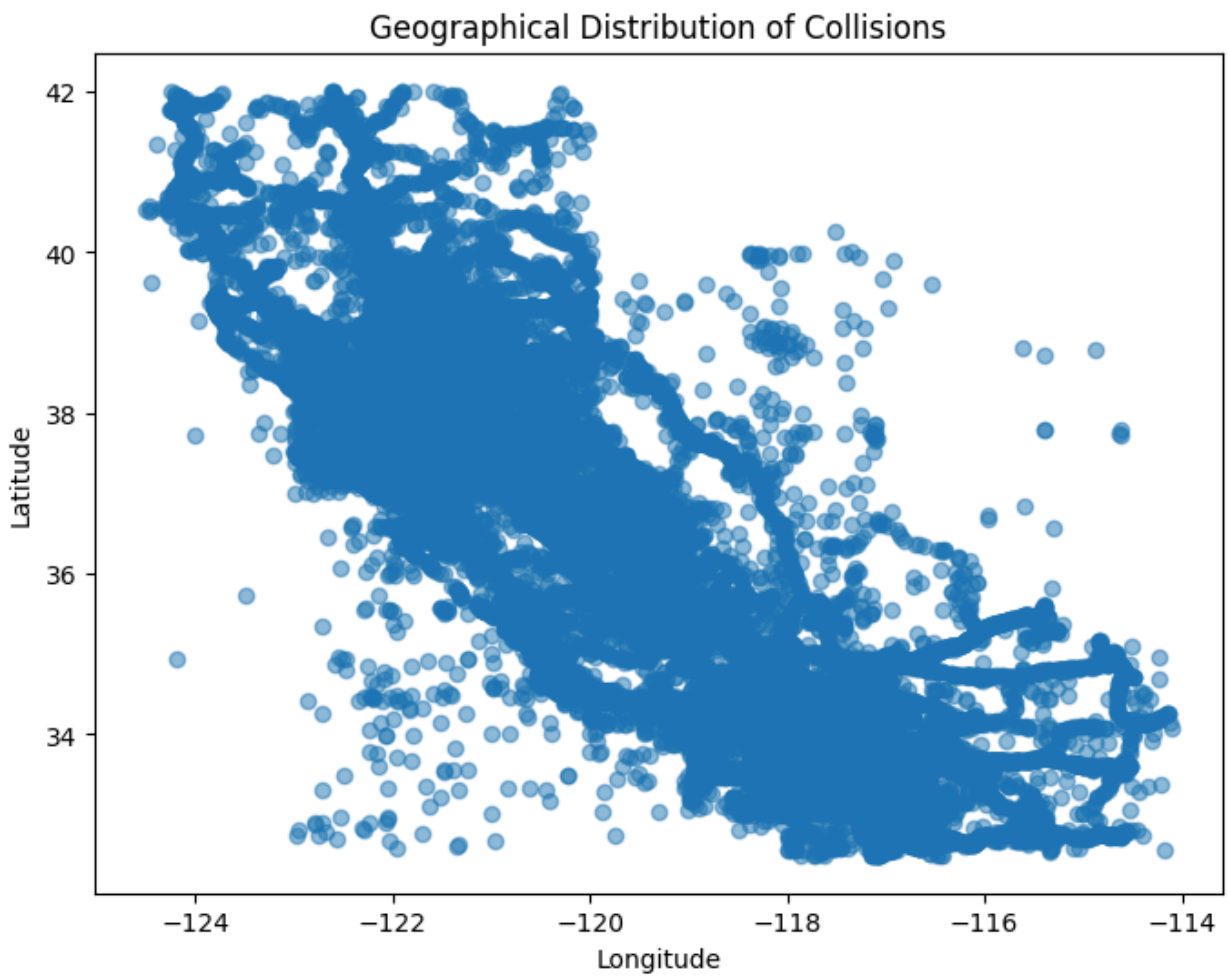
3.1.9. Spatial Analysis # Collision Density by County

I analyzed the spatial distribution of collisions by county using a bar chart. The results show that certain counties like Los Angeles have a significantly higher number of collisions compared to others. This indicates that these regions may have higher traffic density or risk factors, making them important areas for safety improvements.



3.1.10. Scatter Plot of Collision Locations

I analyzed the geographical distribution of collisions using a scatter plot of latitude and longitude values. The data was first cleaned by converting coordinates to numeric format and removing invalid entries. The plot shows clusters of collisions in certain areas, indicating high-density regions where accidents occur more frequently.

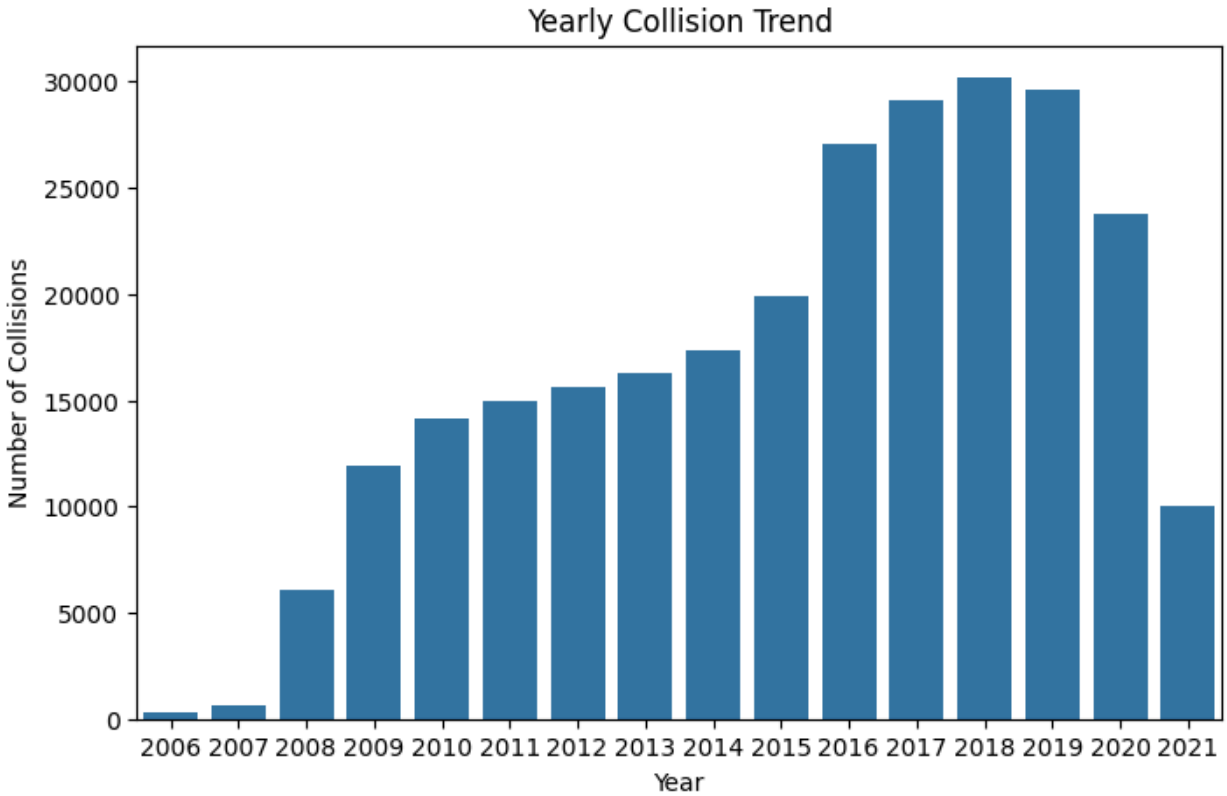


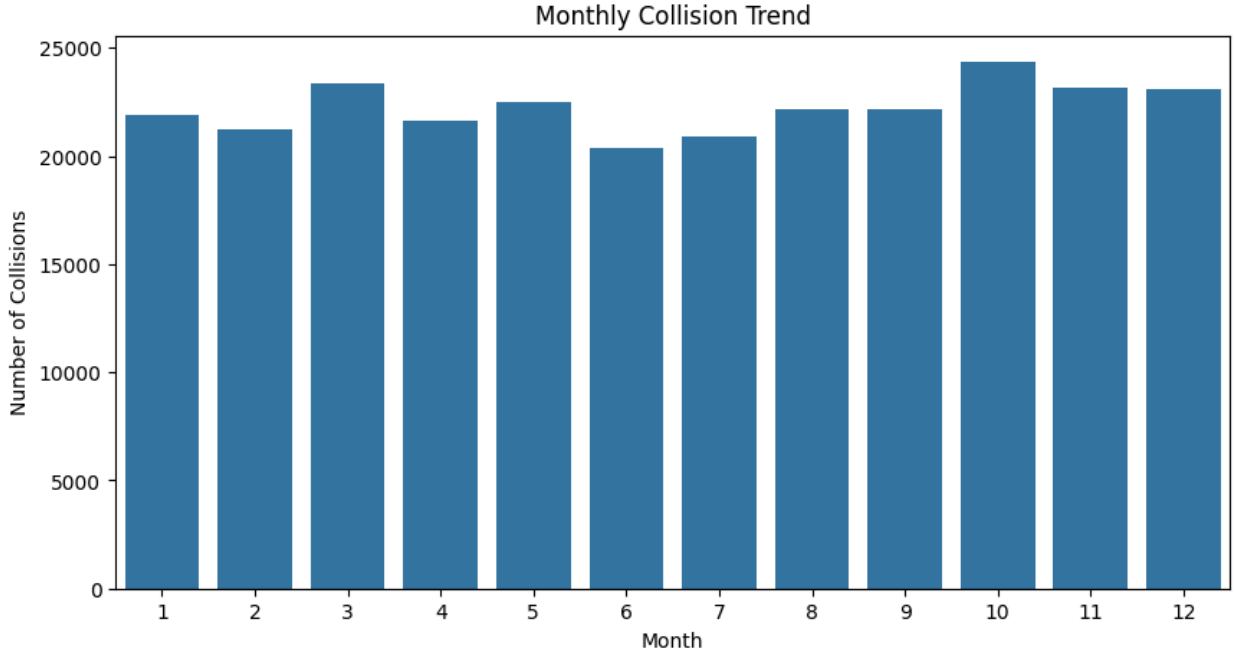
3.1.11. Collision trends over time

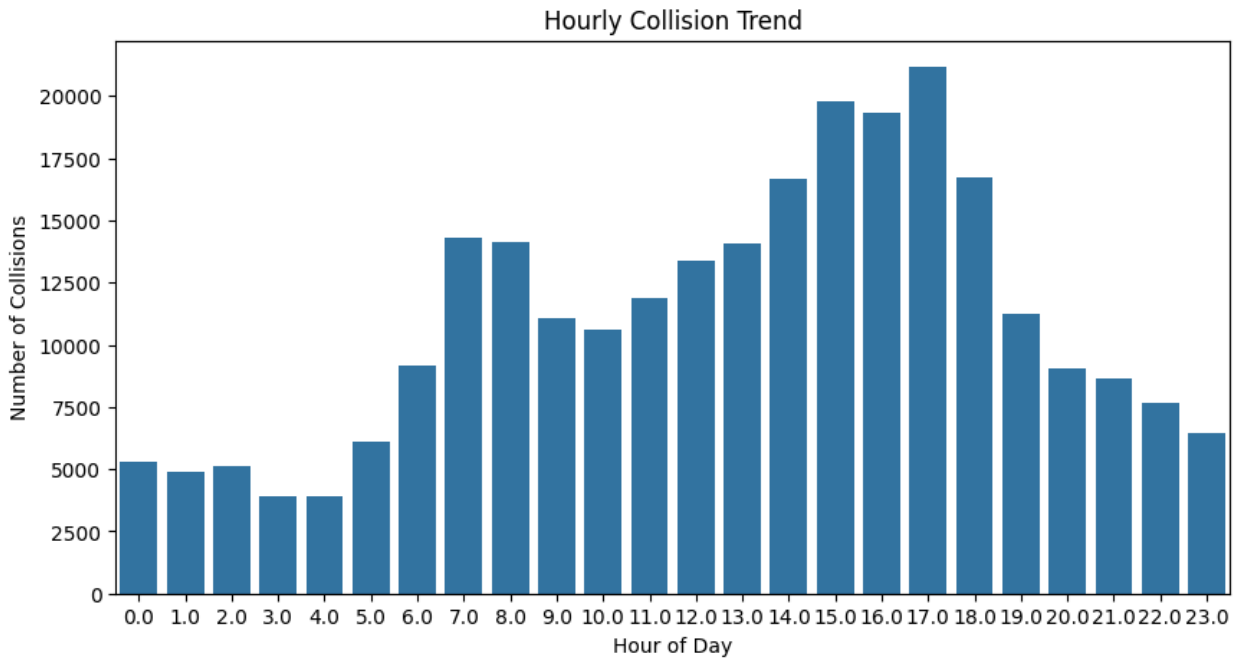
Yearly collision trend - 2018 highest

Monthly collision trend - 10th Month is the highest

Hourly collision trend - 17th hour is the highest


Yearly Collision Trend
Number of Collisions
0
5000
10000
15000
20000
25000
30000
2006 2007 2008 2009 2010 2011 2012 2013 2014 2015 2016 2017 2018 2019 2020 2021
Year


Monthly Collision Trend
Number of Collisions
0
5000
10000
15000
20000
25000
1 2 3 4 5 6 7 8 9 10 11 12
Month



4. ETL Querying

 # Query: Identify the top 5 counties with the most collisions

The top 5 counties with the highest number of collisions are identified by grouping the dataset based on county location and counting the number of collisions. Los Angeles had the highest number of collisions, followed by Orange, Riverside, San Diego, and San Bernardino. This indicates that densely populated and high-traffic regions experience more accidents.

 # Query: Find the month with the highest number of collisions

The analysis shows that October recorded the highest number of collisions, with a total of 24,347 incidents. This indicates a peak in accident frequency during this month, which may be influenced by seasonal changes, increased traffic, or other external factors.

 # Query: Find the most common weather condition during collisions

The most common weather condition during collisions was found to be clear weather, with a total of 219,575 incidents. This indicates that most accidents occur under normal weather conditions, likely due to higher traffic volume and increased road usage during clear weather rather than adverse conditions.

 # Query: Determine the percentage of collisions that resulted in fatalities

The analysis showed that no collisions in the dataset resulted in fatalities, leading to a 0% fatality rate. This is likely due to the dataset being a filtered or sample dataset that does not include fatal cases. In real-world scenarios, fatal collisions do occur but represent a small percentage of total accidents.

 # Query: Find the most dangerous time of day for collisions

The most dangerous time of day was found to be 5 PM (hour 17), with 21,168 collisions recorded. This indicates that evening peak hours are the most accident-prone, likely due to heavy traffic, congestion, and rush-hour conditions.

 # Query: List the top 5 road types with the highest collision frequency

The analysis shows that the majority of collisions occur on dry road conditions, with 241,634 incidents. Wet and snowy conditions contribute to significantly fewer collisions. This indicates that higher traffic volume during normal road conditions leads to more accidents, while adverse road conditions may reduce traffic but increase risk severity.

 # Query: Find the top 3 lighting conditions that lead to the most collisions

The analysis shows that the highest number of collisions occur during daylight conditions, with 175,682 incidents. This is followed by dark conditions with street lights and without street lights. The results indicate that while most accidents occur during the day due to higher traffic volume, poor lighting conditions can contribute to reduced visibility and potentially more severe accidents

5. Conclusion

Final insights and recommendations

5.1 Recommendations to improve road safety by identifying high-risk locations and peak accident times for infrastructure improvements.

From the analysis, it is clear that most collisions occur in highly populated counties such as Los Angeles, Orange, and Riverside. Also, the highest number of collisions happens around 5 PM, which is the evening rush hour. This shows that road safety improvements should focus on high-traffic areas and peak time periods. Measures like better road design, proper lane management, and improved traffic signals during peak hours can help reduce accidents.

5.2 Suggestions to optimize traffic management by analyzing trends in collision severity, weather conditions, and lighting to improve road design and traffic signal timing.

The analysis shows that most accidents happen during clear weather and daylight, mainly because more vehicles are on the road. Traffic management can be improved by using smart traffic signals, better street lighting, and monitoring traffic flow to reduce congestion and accidents.

5.3 Propose data-driven policy changes to enhance pedestrian and cyclist safety based on collision trends involving vulnerable road users.

Even though our provided datasets for this assignment does not directly focus on pedestrians and cyclists, injury-related data shows that accidents still have a serious impact. Safety can be improved by adding pedestrian crossings, dedicated bicycle lanes, and enforcing traffic rules more strictly to protect vulnerable road users.

5.4 Identify potential high-risk zones for proactive intervention by examining geographic collision density and historical accident data.

The spatial analysis shows that certain regions have a higher concentration of accidents. These areas can be treated as high-risk zones. Authorities can focus on these zones by improving road infrastructure, increasing surveillance, and implementing stricter traffic control measures.

5.5 Assess the impact of environmental factors such as weather, road surface conditions, and lighting on accident frequency and severity.

The analysis shows that most accidents occur on dry roads because most driving happens in normal conditions. However, wet and slippery roads can also lead to more severe accidents. Similarly, poor lighting conditions also increase risk due to reduced visibility. This shows that environmental factors may not increase the number of accidents but can affect how serious they are.

5.6 Develop predictive models to anticipate collision hotspots and support proactive safety measures.

Using historical data, it is possible to identify patterns such as high-risk locations and peak accident times. This data can be used to build predictive models that help in identifying accident-prone areas in advance. This allows authorities to take preventive actions like increasing monitoring or improving infrastructure before accidents occur.

Overall Conclusion

The analysis of traffic collision data provides valuable insights into accident patterns across time, location, and environmental conditions. Univariate analysis revealed that most collisions occur under normal conditions such as daylight and clear weather, indicating that traffic volume is a primary driver of accidents. Bivariate analysis showed that factors such as lighting and weather influence accident severity rather than frequency.

Temporal analysis highlighted peak accident hours during evening rush periods, while spatial analysis identified urban counties as high-risk zones. These findings emphasize the importance

of targeted infrastructure improvements, optimized traffic management, and data-driven policy interventions.

Overall, the results demonstrate that effective road safety strategies should focus on:

- Managing traffic density during peak hours
- Improving infrastructure in high-risk regions
- Enhancing visibility and road conditions
- Leveraging data analytics for proactive safety planning

These insights can help policymakers, urban planners, and traffic authorities design safer and more efficient transportation systems.